

Mohamed Sallam

Back-end Engineer

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Work Experience

Master Micro

Part-Time Software Engineer

Sep 2022 - Sep 2023

Full-Time Software Engineer Intern

Jul 2022 - Sep 2022

Worked on the development of an [Analog Designer's Toolbox \(ADT\)](#), a tool found application in thousands of users across USA, China, India, Egypt, and more. It was designed to automate the process of analog circuit design. The excellence of this tool was recognized and appreciated by professors from prestigious universities worldwide.

- Created reusable components for Qt/C++ front-end by performing some algorithms and data structures concepts and using C++ STL functions and containers.
- Applied object-oriented design concepts and design patterns, for example, create a undo/redo behavior for a text editor using *Command Pattern*.
- Learned and performed a bunch of programming principles (eg, KISS, YAGNI, DRY, ...).
- Started new position as a Part-Time Software Engineer and rewarded for **outstanding performance** in the internship.
- Created a fully functional plotting widget using QwtPlot and performed some scientific computation concepts.
- Performed cleaner version control workflow and Learned more intermediate Git topics.

Algorithms Data Structures Object-Oriented Design Design Patterns Debugging Regex JUnit
Git/GitHub Scrum REST APIs Python Doxygen C++ Java CMake Qt Gradle QtTest
PySide2

WRL - Wireless Research Lab, Faculty of Engineering, Helwan University, Cairo, Egypt

Research Developer

Sep 2021 - Aug 2022

Worked on the Laboratory Learning Management System (LLMS) project which is a tool to facilitate the creation of laboratory educational materials, including interactive experiments and quizzes, supported by a grant of over 3 million EGP from the Academy of Scientific Research and Technology in Egypt.

- Created and developed an easy-to-use e-learning materials authoring tool for LLMS using OOP Concepts, Angular, HTML/CSS, Typescript, Angular Material, CkEditor and more.
- Created a model to store these materials in what is known as Laboratory Learning Object (or LLO).
- Led the front-end team, divided the tasks between us and organized the work plan for the development.

Debugging Angular Typescript JavaScript HTML/CSS

Education

Faculty of Engineering, Helwan University, Cairo, Egypt

Sep 2019 - May 2024

- Bachelor of Science in Electronics, Communications and Computer Engineering - Computer and Systems Engineering, grade: 80.05%
- Graduation Project: Lin-K-it, grade: A+

A functional prototype to replace the expensive ELVIS Emona Communications Boards used in communications labs. This software-based solution reduced costs by **34,000%**, making it more accessible to universities and students. The prototype was adaptable for multiple lab environments, such as biomedical engineering, and introduced AI-assisted lab work.

- Conceptualized the idea of a fully computer-based system to replace costly hardware.
- Led the overall system design, coordinating efforts to ensure a cohesive architecture.
- Adapted a Windows-based GUI simulator (written in .NET) to be cross-platform by developing a simple REST-like API using .NET Core.
- Developed the core system to coordinate interactions between the GUI, simulator, and hardware kit, ensuring seamless integration and communication.
- Designed an algorithm for connectivity checking between nodes in the kit, contributing to a cost-effective prototype, though it could not be implemented due to import constraints.
- Designed the user interface and implemented custom sidebar component to navigate between different labs and courses.

Design Patterns FastAPI Python C# .NET Core React.js JavaScript

Projects

Banque of Egypt [↗](#), Java/Spring Boot Bootcamp by EBI

28 Jul 2024 - 15 Aug 2024

A simple CRUD middleware web application that manages transactions between banks, including features for transferring money, registering banks, listing banks, and viewing transaction history. Developed in one day during a 15-day Java/Spring Boot bootcamp by Egyptian Banking Institute.

- Led the team, elicited requirements, assigned tasks, and ensured timely delivery.
- Implemented the transactions feature, including handling failing transactions using the command pattern.
- Wrote the API specifications for other web apps (banks) to be implemented by them.
- Contributed to the design of the database schema.

Design Patterns

Spring Framework

Spring Boot

Spring MVC

Thymeleaf

Spring Data JPA

MySQL

Clean Architecture

Maven

StateLang [↗](#)

25 Apr 2024 - 7 May 2024

A fork of State Machine Compiler for Clean Code video series (CC_SMC) project created by Uncle Bob. SMC is a Java application that translates a state transition table into a program, in C, C++ and Java, implementing the described state machine. The goal of the fork was to enhance SMC by adding support for diagramming languages such as PlantUML and Mermaid.

- Initiated and proposed the project idea and wrote a comprehensive design document outlining the integration of diagramming language support, including PlantUML and Mermaid, along with other programming languages.
- Developed a module to simplify the inheritance tree, optimizing the representation of state machines, utilizing a general tree data structure with a hash map and employing complex set operations and algorithms to eliminate implicit super states, redundant super states, and super states intersection.
- Pair-programmed in the *generator* and *implementer* phases, implementing the visitor pattern and composite pattern.
- Contributed to code reviews and refactored the design to make the CodeGenerator module more abstract, facilitating code-style-agnostic feature.
- Utilized best practices, including pair programming, code review, and studying the codebase to ensure adherence to Uncle Bob's Clean Code principles.

Compiler Design

Java

Design Document

Complex Data Structures and Algorithms

Design Patterns

UML State Diagram

Diagramming Languages

LeetStream [↗](#)

6 Apr 2024 - 15 Apr 2024

A static website aimed at aiding users in enhancing their coding skills through daily practice with LeetCode problems, categorized into 26 coding patterns.

- Implemented my own spaced repetition algorithm to distribute problems from each pattern across spaced repeated periods, optimizing learning retention and effectiveness.
- Engineered a simple, minimalist and efficient routing system along with a lightweight component utility, offering project developers a ReactJS-like coding experience while maintaining simplicity, ease of use, and speed.
- Designed and implemented the entire project from scratch, utilizing web components, plain HTML/CSS, and JavaScript, while adhering to best practices.
- Conceptualized and integrated features such as creating/joining streams, solving daily problems, submitting data for score calculation, reviewing results, accessing study material, and maintaining data privacy.

JavaScript

HTML/CSS

Algorithms

Web Components

Al-Miqata [↗](#)

Apr 2023 - May 2023

Cost-effective, free and open-source digital Islamic wall clock with prayer times

- Translated JavaScript prayer times and Hijri Calendar calculations library into C, implementing embedded-C style and optimizations.
- Modularized software design for enhanced maintainability and scalability.
- Created drivers for seven-segment displays and LEDs multiplexing, as well as basic drivers for RTC, I2C, WIFI, and UART.
- Designed a schematic using cost-effective, locally available components.

C

JavaScript

Modular Programming

Embedded Systems

Circuit Design

Proteus

Awb [↗](#)

Oct 2022 - Jun 2023

Free and Open Source Android App to quit digital addiction

- Created a [prototype](#) of the app and then gathered a team and begun working on the real app.
- Facilitated Scrum meetings, delegated tasks, led and mentored the team from requirement gathering phase (writing [SRS](#)) through design and analysis phase (drawing [CRC cards](#), [ERD](#), [DFD](#), [Use Case Diagram](#), [Class Diagram](#), [Sequence Diagram](#) and

[Wireframes](#)) to [implementation and documentation](#) and submit it as a final project in *Software Engineering Course*.

- Developed many classes and pair-programmed another in pretty all packages.

Object-Oriented Design

Software Engineering and Development

Design Patterns

MVVM Architecture

Git/GitHub

Scrum

Leadership

Android Development

SQL

Volunteer Experience

Resala Charity Organization, Cairo, Egypt

Marketing Volunteer

Jan 2023 – Feb 2023

Languages

English Professional working proficiency

Arabic Native or bilingual proficiency

Interests

EduTech

Coding

Automation

Superuser

Poetry

Logic

Philosophy

Reading